

Day 16 — TNPSC Group 2A Study Material

Part A – General Studies: Indus Valley Civilization

READING MATERIAL

Opening topic of Unit III. Real exam evidence shows two recurring traps: assuming ALL Harappan sites were large urban centres (false — many were small villages), and assuming all major Harappan sites are in India (false — Mohenjodaro and Harappa itself are in present-day Pakistan).

Q: Real exam: True or false — 'All Harappan sites are urban in character.'

A: FALSE. While the major cities (Mohenjodaro, Harappa, Dholavira) were large urban centres, Harappan-culture remains have been found even at small village sites — the civilization depended on villages for food and labour, not just cities.

Q: Real exam: Which major Harappan sites are NOT located in India?

A: Mohenjodaro and Harappa — both are in present-day Pakistan (Sindh and Punjab provinces respectively). Lothal, Kalibangan, Bhanwali, Ropar, Surkotada, and Dholavira are all in India.

Q: What is the most distinctive feature of Lothal?

A: Its dockyard — Lothal (in Gujarat) had a sophisticated dockyard, evidence of the Harappans' maritime/trading capability.

Q: Who discovered the connection between Harappa and Mohenjodaro, establishing them as one civilization, and when?

A: Sir John Marshall, Director General of the Archaeological Survey of India, in 1924 — he found common features between the two sites and concluded they were part of one large civilization.

Q: What were the Indus Valley Civilization's most advanced urban planning features?

A: Baked brick houses, elaborate drainage and sewage systems (among the world's earliest known urban sanitation systems — more advanced than contemporary Middle Eastern cities), and planned water supply.

Q: What famous bronze artifact was found at Mohenjodaro, and what were the Harappan standard weights' ratio?

A: The 'Dancing Girl' bronze figurine. Standard cubical weights were found in the ratio 1:2:4:8:16:32.

Q: Name the five major Harappan urban centres and their general regions.

A: Mohenjodaro (lower Indus, Sindh), Harappa (western Punjab), Ganeriwala (Cholistan desert), Dholavira (Gujarat), and Rakhigarhi (Haryana).

MOCK TEST (WITH ANSWERS)

1. Which statement about Harappan culture is INCORRECT?

- A. All Harappan sites are urban in character ✓**
- B. Harappan cities depended on villages for food and labourers
- C. Harappan remains have been found even at small village sites
- D. Lothal's most distinctive feature is its dockyard

Explanation: Not all Harappan sites were urban — many were small villages. [medium, question-bank-2024]

2. Which of the following major Harappan sites are located in present-day Pakistan, not India?

- A. Lothal and Kalibangan
- B. Mohenjodaro and Harappa ✓**
- C. Surkotada and Dholavira
- D. Bhanwali and Ropar

Explanation: Mohenjodaro and Harappa are both in Pakistan. [medium, question-bank-2024]

3. Which Harappan site is best known for its dockyard?

- A. Mohenjodaro
- B. Harappa
- C. Lothal ✓**
- D. Dholavira

Explanation: Lothal. [easy, question-bank-2024]

4. Who established that Harappa and Mohenjodaro belonged to the same civilization, in 1924?

A. Alexander Cunningham

B. Sir John Marshall ✓

C. Mortimer Wheeler

D. R.D. Banerji

Explanation: Sir John Marshall. [medium, web-research]

5. Harappan standard cubical weights were found in which ratio?

A. 1:2:3:4:5:6

B. 1:2:4:8:16:32 ✓

C. 1:3:9:27:81:243

D. 2:4:6:8:10:12

Explanation: 1:2:4:8:16:32. [hard, web-research]

6. The famous 'Dancing Girl' bronze figurine was found at:

A. Harappa

B. Lothal

C. Mohenjodaro ✓

D. Dholavira

Explanation: Mohenjodaro. [medium, web-research]

Part B – Aptitude & Mental Ability: Number Series

READING MATERIAL

Number Series is the final Reasoning skill. Real exam evidence shows 6 distinct underlying patterns: second-difference (difference-of-differences) sequences, sum-of-consecutive-cubes, geometric progression, arithmetic progression, n^2 multiples, and prime-number-squared sequences — recognize the pattern TYPE before trying to compute term-by-term.

Q: Method — sum of consecutive cubes: Find the next term: 9, 35, 91, ?

A: Each term is $n^3 + (n+1)^3$: $1^3 + 2^3 = 1 + 8 = 9$ ✓, $2^3 + 3^3 = 8 + 27 = 35$ ✓, $3^3 + 4^3 = 27 + 64 = 91$ ✓. Next: $4^3 + 5^3 = 64 + 125 = 189$.

Q: Method — second-difference (arithmetic sequence of differences): Fill in the blank: 1, 6, 15, 28, 45, ____, 91.

A: Differences: $6 - 1 = 5$, $15 - 6 = 9$, $28 - 15 = 13$, $45 - 28 = 17$ — these differences themselves increase by 4 each time. Next difference = $17 + 4 = 21$, so the blank = $45 + 21 = 66$. (Check: $91 - 66 = 25 = 21 + 4$ ✓ consistent.)

Q: Method — sequence of squared primes: Find the missing number in the sequence of squared prime numbers: 4 (2^2), 9 (3^2), 25 (5^2), 49 (7^2), ?, 169 (13^2).

A: The base numbers are consecutive primes: 2, 3, 5, 7, 11, 13. The missing term corresponds to $11^2 = 121$. (Caution: the original source rendered this sequence with garbled formatting that looked like plain 3-digit numbers rather than squared primes — always re-derive the actual pattern rather than trusting a confusing transcription.)

Q: Method — simple arithmetic progression (AP) with increasing-by-2 differences: Find the next term: 1, 5, 11, 19, 29, ?

A: Differences: 4, 6, 8, 10 — increasing by 2 each time. Next difference = 12, so next term = $29 + 12 = 41$.

Q: Method — geometric progression (GP): Find the 8th term of 2, 6, 18, 54, ...

A: Common ratio = 3 (each term $\times 3$). Terms: 2, 6, 18, 54, 162, 486, 1458, 4374. The 8th term is 4,374.

Q: Method — plain arithmetic progression, nth term formula: Find the 20th term of 7, 10, 13, ...

A: First term = 7, common difference = 3. nth term = $a + (n-1)d = 7 + (19 \times 3) = 7 + 57 = 64$.

Q: Method — multiples of n^2 : Find the missing number: 3, 12, 27, 48, 75, 108, ____.

A: Each term = $3 \times n^2$ for $n=1, 2, 3, \dots$: $3 \times 1 = 3$, $3 \times 4 = 12$, $3 \times 9 = 27$, $3 \times 16 = 48$, $3 \times 25 = 75$, $3 \times 36 = 108$. Next ($n=7$): $3 \times 49 = 147$.

MOCK TEST (WITH ANSWERS)

1. Find the next term: 2, 9, 28, 65, 126, ?

A. 215

B. 217 ✓

C. 219

D. 221

Explanation: Each term is $n^3 + 1$: $1^3 + 1 = 2$, $2^3 + 1 = 9$, $3^3 + 1 = 28$, $4^3 + 1 = 65$, $5^3 + 1 = 126$. Next: $6^3 + 1 = 216 + 1 = 217$. [hard, authored]

2. Fill in the blank: 2, 7, 14, 23, 34, ____, 62.

A. 45

B. 47 ✓

C. 49

D. 51

Explanation: Differences: 5, 7, 9, 11 — increasing by 2. Next difference = 13, so blank = $34 + 13 = 47$. (Check: $62 - 47 = 15 = 13 + 2$, consistent.) [medium, authored]

3. Find the next term: 3, 7, 15, 27, 43, ?

A. 61

B. 63 ✓

C. 65

D. 67

Explanation: Differences: 4, 8, 12, 16 — increasing by 4. Next difference = 20, so next term = $43 + 20 = 63$. [medium, authored]

4. Find the 7th term of the GP: 3, 9, 27, 81, ...

A. 1,458

B. 2,187 ✓

C. 2,287

D. 1,987

Explanation: Common ratio 3: 3,9,27,81,243,729,2187. 7th term = 2,187. [medium, authored]

5. Find the 15th term of the AP: 5, 9, 13, 17, ...

A. 59

B. 61 ✓

C. 63

D. 65

Explanation: $a=5$, $d=4$. 15th term = $5+(14 \times 4) = 5+56 = 61$. [easy, authored]

6. Find the missing number: 2, 8, 18, 32, 50, ____.

A. 62

B. 68

C. 72 ✓

D. 78

Explanation: Each term = $2 \times n^2$: $2 \times 1=2$, $2 \times 4=8$, $2 \times 9=18$, $2 \times 16=32$, $2 \times 25=50$. Next ($n=6$): $2 \times 36=72$. [medium, authored]

Part C – General English: Poem: "Advice from a Tree" — Ilan Shamir

READING MATERIAL

A poem using a tree's life as an extended metaphor for life advice — each tree-related image (roots, limbs, seasons, leaves) maps to a piece of wisdom. Unlike most other poems in this syllabus, real exam evidence here favours idiom/phrase MEANING questions over figure-of-speech labelling — know what each phrase means, not just what device it is.

Q: Who wrote "Advice from a Tree", and what's notable about him?

A: Ilan Shamir — notable for having planted over 50,000 trees and led more than 800 tree-planting celebrations worldwide, directly living out the nature-connection theme of his own poem. (He also wrote "The River", elsewhere in this same syllabus.)

Q: What is the poem's central theme?

A: Nature as a guide for living — using a tree's life and growth as a metaphor for how to live well.

Q: What does "Stand tall and proud" mean in the poem?

A: Have confidence and self-respect — not arrogance or literal height, but quiet inner confidence.

Q: What does "Sink your roots deeply into the Earth" mean?

A: Stay grounded and stable — having a firm foundation/values to rely on, just as a tree's roots anchor and stabilize it.

Q: What does "Go out on a limb" mean?

A: Take a risk — a common English idiom (literally, climbing onto a tree branch that might not hold your weight) used here for stepping beyond your comfort zone.

Q: What do the poem's seasons represent, and what does "the wisdom to let go of leaves in the Fall" symbolize?

A: The seasons represent stages and lessons of life. Letting go of leaves in Fall symbolizes letting go of burdens or attachments — knowing when to release what no longer serves you, just as a tree sheds leaves before winter.

MOCK TEST (WITH ANSWERS)

1. "Advice from a Tree" was written by:

- A. Henry Van Dyke
- B. Ilan Shamir ✓**
- C. William Henry Davies
- D. Robert Frost

Explanation: Ilan Shamir. [easy, web-research]

2. The central theme of the poem is:

- A. Urban life
- B. Nature as a guide for living ✓**
- C. Space exploration
- D. Mythical creatures

Explanation: Nature as a guide for living. [easy, web-research]

3. In the poem, "Stand tall and proud" means:

- A. Show off your wealth
- B. Be arrogant
- C. Have confidence and self-respect ✓**
- D. Be physically taller than others

Explanation: Confidence and self-respect. [medium, web-research]

4. "Sink your roots deeply into the Earth" means:

- A. Dig a hole
- B. Stay grounded and stable ✓**
- C. Run away
- D. Become a farmer

Explanation: Stay grounded and stable. [medium, web-research]

5. "Go out on a limb" means:

A. Sit on a tree branch literally

B. Take a risk ✓

C. Cut a tree

D. Be lazy

Explanation: Take a risk — a common idiom. [medium, web-research]

6. "The wisdom to let go of leaves in the Fall" symbolizes:

A. Changing clothes for the season

B. Letting go of burdens or attachments ✓

C. Cutting down trees

D. Buying seasonal gifts

Explanation: Letting go of burdens or attachments. [medium, web-research]